Jalal Ahmmed

Graduate Researcher, Department of Computer Science and Engineering
Daffodil International University, Dhaka, Bangladesh

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Google Scholar LinkedIn GitHub

Education

Bachelor of Science in Computer Science and Engineering (CSE) July 2022 – July 2026
Daffodil International University, Dhaka, Bangladesh
CGPA: 3.78 / 4.00

Higher Secondary Certificate (Science) 2021
Shaheed Ramiz Uddin Cantonment College, Dhaka, Bangladesh
GPA: 5.00 / 5.00

Secondary School Certificate (Science) 2019
Gazirhat High School, Senbagh, Noakhali, Bangladesh
GPA: 4.83 / 5.00

Research Experience

Research Assistant
Kabir Research Group
Department of CSE, Daffodil International University, Dhaka, Bangladesh Feb 2025 – Present
Supervisor: Md. Alamgir Kabir, PhD

- Conducted research on explainable AI and ensemble learning.
- Contributed to manuscript preparation and model evaluation.

Research Interests

• Computer Vision • Explainable Artificial Intelligence (XAI) • Deep Learning • Ensemble Learning • Medical Imaging • Agricultural AI • Trustworthy Machine Learning • Model Interpretability (SHAP, LIME) • Spiking Neural Networks (SNN) • Federated Learning

Selected Research Projects

LBSpikeNet: A Lightweight Hybrid CNN–SNN Framework with Explainable AI for Multi-Brain Disease Classification (BSc Thesis) 2025–2026

Lead Researcher

- Developed a lightweight hybrid CNN–SNN framework for Brain Tumor, Alzheimer's Disease, and Stroke classification.
- Achieved accuracies of **99.24%**, **99.30%**, and **98.00%** across three neuroimaging datasets.
- Integrated **Grad-CAM, Grad-CAM++, LIME, and t-SNE** for model interpretability and clinical trustworthiness.

LBNet: An Optimized Lightweight CNN for Mammographic Breast Cancer Classification with XAI-Based Interpretability 2025

First Author Publication, Scientific Reports (Nature Portfolio)

- Designed a lightweight CNN with only **2.4M trainable parameters** for breast cancer detection.
- Achieved **97.28%** accuracy on RSNA and up to **99.54%** on external datasets.
- Applied **SHAP** and **Grad-CAM** to enhance model transparency and explainability.

HPrEd: A Tuned Ensemble with Model-Agnostic XAI to Explain Social Media's Association with Academic Productivity 2026

First Author Publication, IEEE Access

- Developed a soft-voting ensemble using Extra Trees, k-NN, and MLP models.

- Achieved **98.25% recall** with only **0.53%** probability of false alarm.
- Utilized **SHAP** and **LIME** to identify key factors influencing academic productivity.

Publications

Journal Articles

- **Ahmmed, J.**, Ahmed, F., Kabir, M. A., Ahad, M. T., Jadoon, M. A., Rehman, A. U., & Bermak, A. (2025). *LBNet: An Optimized Lightweight CNN for Mammographic Breast Cancer Classification with XAI-Based Interpretability.* Scientific Reports. DOI: 10.1038/s41598-025-31642-6 (Q1 Journal, Nature Portfolio).
- **Ahmmed, J.**, Kabir, M. A., Rehman, A. U., & Bermak, A. (2026). *A Lightweight Hybrid CNN and Transformer Model for Medicinal Leaf Disease Classification with Explainable AI.* Scientific Reports. DOI: 10.1038/s41598-026-39182-3 (Q1 Journal, Nature Portfolio).
- **Ahmmed, J.**, Ahmed, F., Kabir, M. A., Ali, N., Rehman, A. U., & Bermak, A. (2026). *HPrEd: A Tuned Ensemble with Model-Agnostic XAI to Explain Social Media's Association with Academic Productivity.* IEEE Access. DOI: 10.1109/ACCESS.2026.3663639 (Q1 Journal, IEEE).

Conference Papers (Presented)

- **Ahmmed, J.**, Ema, J. N., Kabir, M. A., Uddin, A. Z. M. J., & Bhuiyan, T. (2025). *Impact of Feature Selection on Classification: Comparing Chi-Square and PCA*. Proceedings of the 5th International Conference on Applied Intelligence and Informatics (AII 2025), Washington, D.C., USA, 28–30 October 2025. (Springer CCIS Proceedings, Scopus Indexed).
- **Ahmmed, J.**, Arafat, Y., Islam, M. N., Sangit, S., Rana, M. M., & Islam, M. M. (2025). *XCLBoost: Soft Voting Ensemble and Explainable AI for Modeling Student AI Adoption.* Proceedings of the 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Dhaka, Bangladesh, 25–27 September 2025.
- **Ahmmed, J.**, Rana, M. M., Sarower, A., Arafat, A., Islam, M. N., & Hossain, I. (2025). *HybridRELR: A Model-Agnostic XAI-Driven Stacking Ensemble for Detecting Financial Fraud.* Proceedings of the 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Dhaka, Bangladesh, 25–27 September 2025.
- **Ahmmed, J.**, Ahmed, F., Kabir, M. A., Zahid, A., Uddin, M. J., & Bhuiyan, T. (2025). *RVE: An Optimized Ensemble Model for Breast Cancer Detection.* Proceedings of the 5th International Conference on Trends in Electronics and Health Informatics (TEHI 2025), Samarkand State University, Samarkand, Uzbekistan, 7–8 November 2025.
- Ahmed, F., Hasan, M., Hossain, S., Ahmed, F., Hasan, M. K., **Ahmmed, J.** (2025). *DRX Ensemble: An Optimized Rank-Based Deep CNN Model for Java Plum Leaf Disease Classification.* Proceedings of the 16th IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT 2025), Indian Institute of Technology (IIT) Indore, Madhya Pradesh, India, 6–11 July 2025.
- Rana, M. M., **Ahmmed, J.**, Kabir, M. A., Zahid, A., Uddin, M. J., & Bhuiyan, T. (2025). *EPX-HDC: An Explainable Gradient-Boosting Model for Heart Disease Classification Using LIME.* Proceedings of the 5th International Conference on Trends in Electronics and Health Informatics (TEHI 2025), Samarkand State University, Samarkand, Uzbekistan, 7–8 November 2025.

Preprint

- **Ahmmed, J.**, Ahmed, F., Shohan, R. H., Rana, M. M., & Hasan, M. (2025). Fine-Tuned CNN-Based Approach for Multi-Class Mango Leaf Disease Detection. *arXiv preprint arXiv:2510.05326*, Computer Vision and Pattern Recognition (cs.CV). DOI: 10.48550/arXiv.2510.05326

Published Datasets

- **Ahmmed, J.**, Ahmed, F. (2026). Dataset on Social Media Usage, Academic Performance, and Productivity Behaviors among Students. *Mendeley Data*, V2. [Go to Dataset]
- **Ahmmed, J.**, Kabir, M. A., Rehman, A. U., Bermak, A. (2026). Medicinal Plant Leaf Disease Dataset. *Mendeley Data*, V2. [Go to Dataset]

Submitted and Ongoing Research Projects

- **LightHCF: Lightweight Hybrid CNN–Transformer with Explainable AI for Multiclass Acute Lymphoblastic Leukemia Classification** — Submitted to *IEEE Open Journal of the Computer Society.*
- **Hybrid CNN–SNN Framework for Lung Cancer Detection** — Ongoing research on lightweight explainable deep learning for lung cancer classification; manuscript in preparation for *IEEE Transactions on Medical Imaging.*

Editorial & Review Activities

- Reviewer, *Education and Information Technologies* (Springer, SCI-indexed, Q1)
- Reviewer, *IEEE Open Journal of the Computer Society* (IEEE, SCI-indexed, Q1)
- Reviewer, *International Journal of Computational Intelligence Systems* (Springer, SCI-indexed, Q1)
- Review Committee Member, *International Conference on Electrical and Computer Engineering Research (ICECER 2026)*, Topkapi University, Istanbul, Türkiye

Open-Source Research Software

- **HPrEd**: A tuned ensemble learning framework with SHAP and LIME-based explainability for analyzing the relationship between social media usage and academic productivity. [Repository]
- **LSeTNet**: A lightweight hybrid CNN–Transformer architecture with Explainable AI for medicinal leaf disease classification and agricultural decision support. [Repository]
- **Bangla Suspicious Text Detection**: A CNN–BiLSTM–Attention based framework for abusive and suspicious Bangla text classification with custom preprocessing and feature extraction. [Repository]
- **Intrusion Detection System (CIC-IDS2017)**: A network intrusion detection framework using Random Forest and Extra Trees ensemble learning with LIME-based model interpretability. [Repository]

English Proficiency

IELTS Academic 16 September 2026
Overall Band Score: **6.5**
Listening: **6.5** – Reading: **7.0** – Writing: **7.0** – Speaking: **6.0**
CEFR Level: **B2**

Technical Skills

- **Programming Languages:** Python, C/C++, SQL, JavaScript, PHP
- **Artificial Intelligence & Machine Learning:** Computer Vision, Deep Learning, Explainable AI (XAI), Ensemble Learning, Transfer Learning, Medical Image Analysis, Agricultural AI, Natural Language Processing (NLP), Federated Learning, Spiking Neural Networks (SNN)
- **Deep Learning Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn
- **Explainable AI Techniques:** SHAP, LIME, Grad-CAM, Grad-CAM++, t-SNE, Feature Importance Analysis
- **Research Methodology:** Experimental Design, Dataset Development, Data Augmentation, Ablation Studies, Cross-Validation, Hyperparameter Optimization, Performance Evaluation, Statistical Analysis
- **Scientific Writing & Publishing:** LaTeX, Overleaf, Academic Writing, Literature Review, Manuscript Preparation, Journal Submission, Peer Review, Research Documentation, Citation Management (Mendeley)
- **Research Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, Kaggle, Mendeley, Microsoft Office Suite
- **Software Development & Deployment:** Flutter, Laravel, REST API Integration, Web-Based ML Applications, Model Deployment
- **Research Competencies:** Scientific Communication, Academic Presentation, Research Mentoring, Project Coordination, Interdisciplinary Collaboration

References

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Dr. Arif Mahmud
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